

		Hence the amplitude of the oscillations is 15 cm.
18	C	After passing through the first polariser, intensity is halved, i.e. 20 W m^{-2}. Subsequently, the intensity after passing through the 2nd and 3rd polarisers is $I_f = \frac{1}{2} I_0 \cos^2 \theta \cos^2 (90^\circ - \theta) = \frac{1}{2} I_0 \cos^2 \theta \sin^2 \theta$ $\cos \theta \sin \theta = \sqrt{\frac{2I_f}{I_0}}$ $\sin 2\theta = 2\sqrt{\frac{2I_f}{I_0}} = 2\sqrt{\frac{2 \times 2.5}{40}} = \frac{1}{\sqrt{2}}$ $\theta = 22.5^\circ$

Raffles Institution
Year 5-6 Physics Department

19	R	n²
-----------	----------	----------------------